

AI Enhanced Personal Knowledge Base

Project Details

Role: Software Developer

Project Type: ai_ml

Difficulty: junior

Duration: 365 days

Focus: api_development, authentication, database, fullstack, problem_solving, testing, security, system_design, ai_integration, frontend, automation, data_visualization

Student Name: Rahul Sharma

Description: A full stack web app where users store, tag, and retrieve notes, with an AI service that can summarize selected notes. Includes JWT role based auth, PDF export, and a simple LangChain Groq integration.

Technologies

- JavaScript
- TypeScript
- React
- Next.js
- Node.js
- Express.js
- Laravel (PHP)
- Tailwind CSS
- Python
- MySQL
- REST API
- GraphQL
- LangChain
- NumPy
- Groq API

Requirements

- Implement a secure, full stack note taking app with AI summarization.
- Use candidate's existing skills and introduce minimal new concepts.

Objectives

- Create JWT based auth with role based access.
- Implement CRUD for notes with tagging.
- Integrate a Python AI microservice for summarization.
- Provide PDF export of notes.
- Write unit, integration, and e2e tests.
- Deliver documentation and deployment scripts.

Functional Requirements

- Users register/login and receive a JWT.
- Admins can view all users' notes; regular users see only theirs.
- Create, read, update, delete notes with title, content, tags.
- Search notes by keyword or tag.
- POST /ai/summary returns a concise summary of given note text.
- Export a note as a downloadable PDF.

Non Functional Requirements

- Passwords stored with bcrypt hash.
- All API responses under 500 /ms for CRUD operations.
- Input data validated server side.
- JWT signed with strong secret and short expiry.
- Basic request logging and error handling.

Modules

Auth

Description: Handles registration, login, JWT issuance, and role checks.

Features

- User model
- Password hashing
- Login endpoint
- Role middleware

Notes API

Description: CRUD operations for notes and tag management.

Features

- Note schema
- Tag schema
- Search endpoint
- Authorization checks

AI Service

Description: Python microservice using LangChain + Groq to summarize text.

Features

- FastAPI wrapper
- Summarization prompt
- Rate limiting

Frontend UI

Description: Next.js + React interface for note management and AI features.

Features

- Dashboard
- Editor with tag picker
- Summary view modal
- PDF export button

PDF Export

Description: Server side generation of PDF from note content.

Features

- HTML to PDF conversion
- Download endpoint

Implementation Plan

Phase 1 – Planning

Tasks

- Gather detailed requirements
- Design ER diagram and API spec

Phase 2 – Backend Core

Tasks

- Set up Node/Express project
- Implement Auth module with JWT
- Create MySQL schema

Phase 3 – Notes & PDF

Tasks

- Develop Notes API
- Add search and tag endpoints
- Implement PDF export route

Phase 4 – AI Integration

Tasks

- Build Python FastAPI service
- Connect LangChain to Groq API
- Create proxy endpoint in Node

Phase 5 – Frontend

Tasks

- Scaffold Next.js with Tailwind
- Build pages for login, dashboard, editor
- Integrate AI summary modal

Phase 6 – Testing & Delivery

Tasks

- Write unit & integration tests
- Add Cypress e2e tests
- Prepare README, API docs, Docker Compose

Evaluation Criteria

Correctness

Description: All functional requirements work as specified.

Weight: 20

Code Quality

Description: Clean, modular code with proper TypeScript typings.

Weight: 20

Architecture

Description: Logical separation of concerns and clear API design.

Weight: 20

Testing

Description: Comprehensive automated test coverage.

Weight: 15

Security

Description: Proper password hashing, JWT handling, and input validation.

Weight: 15

Documentation & Deployability

Description: README, API docs, and Docker Compose work out of the box.

Weight: 10

Deliverables

- Source code repository
- README with setup instructions
- OpenAPI specification
- Database schema diagram
- Automated test suite
- Docker Compose file for full stack deployment

Api Endpoints

POST

Path: /api/auth/register

Purpose: Create new user

POST

Path: /api/auth/login

Purpose: Authenticate and return JWT

GET**Path:** /api/notes**Purpose:** List user's notes with optional filters**POST****Path:** /api/notes**Purpose:** Create a new note**POST****Path:** /api/ai/summary**Purpose:** Return AI generated summary of supplied text**GET****Path:** /api/notes/:id/pdf**Purpose:** Download note as PDF

Student Fit: Rahul already knows JavaScript, TypeScript, React, Next.js, Node, Express, JWT auth, and MySQL basics, so the core stack matches his skill set. The project adds a modest Python microservice for AI, introducing LangChain and Groq API without overwhelming complexity, fitting a junior level but allowing a year long timeline for deeper learning and thorough testing.

Student Id: 6ab2576039d4e38828d96a4e

Status: active